

RES-5 all-order common-mode bound: the closure reduces to a geometric

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

The RES-5 entry note established the common-mode control parameter a_0 and the leading envelope. The operator directed the all-order bound (three goals: the all-order split; the per-order $O(a_0)$ bound; the summed series with $C_{\text{cm}} < 10.4$). This note executes the programme as a REDUCTION: it shows RES-5 closure is equivalent to a geometric-domination condition on the condensate-perturbation series, and that the base (measured) order sits a factor ~ 5 inside the closure region. It does NOT close RES-5 – the all-order domination is the deep residual. Scope: the structure and the closure arithmetic; anchor μ^2 , operating endpoint.

2. The condensate-perturbation series (goal 1: the split)

At fixed I the dressing $\hat{r}(I)$ is pattern-independent, so the exact selection difference is a perturbation series in the condensate $\lambda' \mathcal{P}^2 = O(a_0) \hat{r}$ on the common sea $D_0(I)$:

$$\Delta F = \sum_{k \geq 2} \Delta F^{(k)}, \quad F^{(n)} = \underbrace{F_{\text{cm}}^{(n)}(D_0)}_{\text{cancels in } \Delta F^{(n)}} + F_{\text{pd}}^{(n)}(D_0; \mathcal{P}^2), \quad \Delta F^{(k)} \sim a_0^{k-1} \chi^{(k)},$$

where $\chi^{(k)}$ is the k -th connected density susceptibility of the common sea (pattern-independent). The common-mode part $F_{\text{cm}}^{(n)}$ – which carries the STRONG Brazovskii fluctuations – cancels exactly in the difference; the pattern-dependent part enters with one power of a_0 per order.

3. The per-order ratio and the closure condition (goals 2–3)

Define the geometric ratio

$$r_k = \frac{\Delta F^{(k+1)}}{\Delta F^{(k)}} = a_0 \frac{\chi^{(k+1)}}{\chi^{(k)}}.$$

Reduction. If $r_k \leq r < \frac{1}{2}$ for all $k \geq 2$, the geometric tail gives

$$\left| \sum_{k \geq 3} \Delta F^{(k)} \right| \leq \Delta F^{(2)} \frac{r}{1-r} < \Delta F^{(2)} \implies \Delta F_{\text{exact}} = \Delta F^{(2)} \left(1 - \frac{r}{1-r} \right) > 0,$$

so RES-5 closes. The SUFFICIENT condition is $r < \frac{1}{2}$ (at $r = 0.49$ the tail is $0.96 < 1$; at $r = 0.51$ it is $1.04 > 1$). In the operator's form this is $C_{\text{cm}} a_0 < 1$ with $C_{\text{cm}} = \frac{1}{a_0} \cdot \frac{r}{1-r}$, i.e. $C_{\text{cm}} < 1/a_0 = 10.5$.

4. The base margin

The base ratio is $r_2 = a_0 \chi^{(3)}/\chi^{(2)}$. The susceptibility ratio is $O(1)$ (SC-SCOPE put the $n = 3$ endpoint effect at a few percent, consistent with $r_2 \sim a_0$, far from $\frac{1}{2}$). Numerically,

$$r_2 \sim a_0 = 0.096 \quad (\text{factor} \sim 5 \text{ below } \tfrac{1}{2}), \quad C_{\text{cm}} \sim O(1) \quad (\text{factor} \sim 10 \text{ below } 10.5),$$

and, IF $r_k \leq a_0$ at all orders, $\Delta F_{\text{exact}} = \Delta F^{(2)}(1 - 0.106) = 0.894 \Delta F^{(2)} > 0$. The condensate-perturbation series starts well inside its convergence radius.

5. Status and the residual

T3 PROOF SKETCH. Established: the condensate-perturbation structure ($\Delta F^{(k)} \sim a_0^{k-1}$, common-mode cancellation), the geometric closure condition $r < \frac{1}{2}$ ($\Leftrightarrow C_{\text{cm}} < 10.5$), and the base margin ($r_2 \sim a_0$, factor ~ 5 inside). **OPEN (the deep residual):** the ALL-ORDER geometric domination $r_k \leq r < \frac{1}{2}$, i.e. the Brazovskii-sea susceptibility ratios $\chi^{(k+1)}/\chi^{(k)}$ stay bounded as $k \rightarrow \infty$. This is a strong-coupling all-order statement: the common-mode cancellation removes the strong COMMON fluctuations, but the pattern-dependent susceptibility ratios at high order are not bounded here. No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **B1 closure** $\Leftrightarrow$ this all-order domination.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*You reduced but did not prove the all-order domination.*” VALID, by design (T3): the structure, the closure condition, and the base margin are established; the all-order χ -ratio bound is the residual – a genuine strong-coupling all-order result, not closeable by the present elementary means.
- (β) “ *$r_2 \sim a_0$ assumes $\chi^{(3)}/\chi^{(2)} = O(1)$.*” VALID-with-mitigation: SC-SCOPE’s $n = 3$ thin endpoint (a few percent) supports $r_2 \sim a_0 \sim 0.1$, but that is the $n = 3$ datum, NOT a proof that $\chi^{(k+1)}/\chi^{(k)} = O(1)$ for all k . The base margin is real; the extrapolation is the residual.
- (γ) “*Strong Brazovskii fluctuations could blow up the high-order χ -ratios.*” VALID and UNSOLVED: this is precisely the residual. The common-mode cancellation is the reason this is even plausible – it removes the strong common part – but the pattern-dependent ratios are unproven at high order.

Quantitative sanity check (mandatory): *threshold* – $r/(1-r) < 1 \Leftrightarrow r < \frac{1}{2}$ (at $r = 0.49 \rightarrow 0.96$, $r = 0.51 \rightarrow 1.04$); *base* – $r_2 \sim a_0 = 0.096$, factor 5.2 below $\frac{1}{2}$; *operator form* – $C_{\text{cm}} < 1/a_0 = 10.5$, structural $C_{\text{cm}} \sim O(1)$; *envelope* – $1 - a_0/(1 - a_0) = 0.894 > 0$ at the base ratio.

7. Result footer

Result ID: B1-RH-ENUM / RES-5 all-order common-mode reduction
Precise statement: The condensate-perturbation series $\Delta F = \sum_k \Delta F^{(k)}$, $\Delta F^{(k)} \sim a_0^{k-1} \chi^{(k)}$, has the common-mode (strong-fluctuation) part cancel in the difference. RES-5 closes iff the geometric ratio $r_k = \Delta F^{(k+1)}/\Delta F^{(k)} = a_0 \chi^{(k+1)}/\chi^{(k)}$ satisfies $r_k \leq r < 1/2$ for all k ($\Leftrightarrow C_{\text{cm}} < 1/a_0 = 10.5$). The base $r_2 \sim a_0 = 0.096$ is a factor

~ 5 below $1/2$; $C_{cm}(1)$ is a factor ~ 10 below 10.5 . IF the all-order domination holds, $\Delta F_{exact} \geq 0.894 \Delta F^{(2)} > 0$.

Scope: Structure + closure arithmetic; anchor μ^2 , endpoint. The all-order domination is OPEN.

Dependencies: res5-commonmode-envelope (a0 control parameter); neargap-common-mode-resolution (R-U10-3); SC-SCOPE (n=3 base); Math424-AddA (rR, lam').

Evidence grade: ANALYTIC (structure + reduction) + EXECUTED (res5_allorder_commonmode.py 4/4).

Reproduction command: python codes/vacuum/res5_allorder_commonmode.py

Expected output: a0=0.096; $r < 1/2$ threshold; base $r_2 \sim a_0$ (x5 below); $C_{cm} < 10.5$; envelope $0.894 > 0$; 4/4.

Falsification gate: A chi-ratio $\chi^{(k+1)}/\chi^{(k)} \geq (1/2)/a_0 \sim 5.2$ at some k (would push $r_k \geq 1/2$ and break the geometric closure); or a non-perturbative-in-condensate contribution evading the a0 power counting.

Tier before / after: No tier flip. B1 T6 on {H-LAYER} unchanged. RES-5/GAP-2: reduced to all-order geometric domination $r_k < 1/2$; base a factor ~ 5 inside.

No-overclaim statement: This does NOT close RES-5/GAP-2. Only the closure CONDITION (geometric domination $r < 1/2$) and the BASE margin ($r_2 \sim a_0$, factor ~ 5 inside) are established; the all-order susceptibility-ratio bound is the deep residual (a strong-coupling all-order statement). B1 stays T6 on {H-LAYER}.

Next required action: Bound the Brazovskii-sea susceptibility ratios $\chi^{(k+1)}/\chi^{(k)}$ (the all-order geometric domination), e.g. via a bubble-iterate / spectral bound on the dressed density correlations; then RES-5 $\rightarrow$ controlled-error theorem and H-LAYER's last deep axis is controlled.